



Excuses, excuses: moral agency and the professional identity of AI developers

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Abstract

Artificial intelligence developers, machine learning engineers, and data scientists occupy a contradictory role in the modern marketplace. While they are central to the business and science of AI, they are marginalized as moral agents. Consequently, the marketplace has cultivated environments in which developers can be unthinking in their own roles and responsibilities, while at the same time tasking them with creating “thinking machines.” The central aim of this article is to show that this state of affairs is morally unjustifiable. To accomplish this, we draw from Arthur Isak Applbaum’s work on adversary roles and Alasdair MacIntyre’s framework for professional moral agency to establish the context dependencies for a “good” AI developer. We then draw from available studies that have engaged developers in questions about their moral agency and place them in conversation with Dennis Thompson and Helen Nissenbaum about the excuses associated with “the problem of many hands,” a concept that has beguiled accountability in the AI community for decades. We then return to MacIntyre’s framework to provide evidence from the same set of studies that AI developers do understand themselves as being responsible for more than just the role, yet they lack a robust community to whom they can submit their choices for ethical scrutiny and work environments that are often non-conducive to their moral actualization. We conclude with specific recommendations for bringing developers’ moral agency more fully into the discourse about AI ethics.

Keywords AI ethics · Professional ethics · Moral agency · The problem of many hands

1 Introduction

Developers of artificial intelligence (AI), data science (DS), and machine learning (ML) systems represent a uniquely empowered subset of skilled practitioners in the modern marketplace. The experts occupying these roles are building, deploying, and maintaining algorithms that are increasingly used to make or support decision-making in nearly every aspect of human life. This includes agriculture, healthcare,

education, communications, manufacturing, supply chain, policing, finance, transportation, and warfare. Developers have a disproportionate level of influence over the flow of information and the technical standards that now shape many of our social interactions. The choices they make about system design and deployment can affect the economic opportunities of millions of people, positively but also very negatively. With every deployment, across every industry, publics around the world become more dependent upon (and defenseless against) developers’ skills.

Yet, dependence on the expertise of others is not a new human experience. We are accustomed to being dependent on surgeons, pilots, lawyers, and bus drivers, among others. Our growing dependence on developers, however, is different in three morally relevant ways. These have to do with proximity, scale, and the ability to regulate the practice. First, with respect to proximity, developers rarely (if ever) come into meaningful contact with the publics their systems affect. Developers can, and often do, work in isolation from those who end up using their systems or who may be affected by them. Howard and Gugger (2020)

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emphasize this point when they advise developers that “the product managers, executives, judges, journalists, doctors—whoever ends up developing and using the system of which your model is a part—will not be well-placed to understand the decisions you made, let alone change them” (p. 179). Compounding this matter is that developers make substantially all the decisions about a model while it is in development, often independently. This includes deciding where to obtain the data, how to organize or transform those data, and which models, predictors, and optimization targets to use (Griffin et al. 2023). These decisions can all be made without speaking to or seeing anyone in the data set or in the population that the system will affect.

By contrast, the pilot, the bus driver, the surgeon, and the lawyer all come into direct contact with the people who rely on their expertise. There is an intimacy to the encounter, fleeting as it may sometimes be, between the dependent person and these specialists. The pilot and the bus driver know there are people sitting behind them who just want to arrive at their destination safely; they may have even welcomed their passengers personally. The surgeon comes into direct physical contact with her patient and will have to talk to both the patient and the family after the surgery. Lawyers must meet with their clients, understand their needs, and sit next to them during court proceedings. Developers do none of these things. They may know at an intellectual level that their model will, for example, wrongly identify some people as having committed fraud, but they do not sit with the accused or talk to their families. Developers are anonymous to the public and the public is anonymous to the developer.

Second, when our lives are in the hands of pilots, bus drivers, lawyers, and surgeons, the scale of potential harm is relatively limited. Catastrophic failures may still occur, which can result in people being severely harmed, and, in the case of a plane crash, even several hundred lives being lost. We make this point not to devalue the very real and devastating consequences of encounters with negligent or nefarious practitioners. Rather, we do so to emphasize that there is a limit to how much damage a person in one of these roles can do, either knowingly or unknowingly. In contrast, AI developers are working at scale. An algorithm designed for use by the Dutch government, for example, wrongly accused approximately 26,000 families of tax fraud, which led to exorbitant financial penalties that drove innocent people into poverty and some to suicide; more than a thousand children were placed in foster care (Heikkilä 2022). It is not merely the AI developers who are to blame in this example, of course. The public officials, judges, and social workers who mobilized against the purported fraud are also responsible. But these Dutch officials were empowered to take the actions they did *because* of the algorithm. The developers are deeply implicated in the harm caused to tens of

thousands of people, yet we can find no evidence that they have been asked to account for their choices.

Third, because of the interpersonal (even if limited) relationship between pilots, surgeons, bus drivers, lawyers and the people they serve, the public’s ability to engage with these professionals as moral agents is rather straightforward. It is much easier, for example, to establish rules, licenses, and systems that govern these practitioners’ work. Developers’ relationship with the public is far more bedeviling. They do not need a license to practice. There are no professional rules or recognized standardization bodies that bind their conduct. They do not self-regulate, nor do they profess as a group to protect the public good. In the popular imagination, they are more often viewed as entitled “tech bros” than as professionals doing morally significant work.

AI developers thus occupy a professionally contradictory role. They are central to the business and science of AI yet marginalized as moral agents. As a consequence, we have cultivated environments in which AI developers can be unthinking in their own roles and responsibilities, all while being asked to create “thinking machines.” In this paper, we submit this state of affairs to a philosophical critique. We will show that the status quo is morally unjustifiable and that the agency of AI developers needs to figure more prominently in our discussions about ethical AI.

We begin with an examination of what it means to be a moral agent at work and what a “good” AI developer might look like. For this, we rely on work by Applbaum (1999) and MacIntyre (1999), who have both written at length about professional moral identity. The outcome of this analysis will lead us into an evaluation of the “problem of many hands,” which has long plagued the question of moral responsibility in software engineering. For this evaluation, we will engage with Thompson (1980) and Nissenbaum (1996) to describe four common excuses used in “many hands” arguments and show how each fails to adequately exempt developers from the discussions about moral responsibility. It will then be necessary to ask whether (and if so, how) AI developers perceive themselves as moral agents. We thus look to the available literature that has engaged with developers to understand whether and how their experiences align with MacIntyre’s moral agency framework. In the concluding section, we will return to Applbaum and argue that because AI developers are morally entangled in the consequences of their creations, they should engage (and be engaged with) more seriously in our efforts to ensure ethical AI.

2 Moral agency and the ‘good’ AI developer

When we discuss “morality” in this article we refer to it in the commonly understood way as pertaining to good or bad behavior, which flows from good or bad character traits.

Moral action, therefore, comports with what is considered appropriate or inappropriate behavior. In this commonly understood view, to be a moral agent is to have the capacity to engage in moral reasoning to thus be held accountable for one's actions (Himma 2009). Moral agents are subject to moral standards of right and wrong conduct, and they can rationally explain their choices on these grounds.

Yet, when a moral agent steps into an occupational role, their agency can become compromised. This is because the activities expected by an occupational role can sometimes be counter to what would be acceptable outside of that role. You would not normally be permitted to access your neighbor's online search history and geolocation, for example. It would be a violation of your neighbor's privacy and deeply disturbing besides. But if you were acting in the role of an AI developer, you might be permitted (indeed, expected) to access the online activities of all your neighbors, *and* to deploy an algorithm that iteratively manipulates their purchasing habits.

Applbaum (1999) called these kinds of jobs “adversary roles,” which he characterizes as those that lay claim to the use of deception and coercion in the service of alleged good ends. His primary examples are politicians (who can willfully distort their opponents' records and thus deceive the public), business managers (who can exploit regulatory loopholes or sell products they know to be useless), and lawyers (who can use practices that may prevent justice from being served). The moral justifications for these acts are numerous, but commonly involve appeals to implied consent (everyone expects a politician to lie, therefore the public consents to being deceived), the fallacy of futility (if I do not coerce and deceive the public for my good ends someone else will, and that someone else's ends are much worse than mine), and the rules of fair play (if the rules of business, politics, and justice permit some level of deception and coercion, it is only fair to allow all the players to coerce and deceive). Applbaum is critical of these justifications because their logical end is to give cover to moral agents for immoral acts.

We argue that AI developers are also in adversarial relationships with the public by virtue of the practices of surveillance and manipulation that are so often core to the work. As just one example, AI developers have been granted broad permission by their employers to build and deploy automated decision-making/support products that can learn from data. To succeed in this endeavor, the developers require a great deal of said data. The developers who worked on the first generative models, for example, had role-prescribed reasons to take as much data as they could from the internet to train their models. They did this without regard for the rights of the content owners. This may make the developers “good” in the technical and role-related sense of having completed

the tasks assigned to them, but they nonetheless procured the intellectual and legal property of others without consent.¹

This act constitutes what Applbaum calls an “unjustified person violation,” which he defines in more or less Kantian terms as actions that damage or deny a person's capacity to choose their own ends (p. 142). Applbaum rejects the argument that the moral responsibility for such violations fall on the role, not on the person, claiming, “A role that requires *morally unjustified* person violation[s] is not a morally justified role, and one cannot acquire a moral obligation to practice it” (p. 247, emphasis ours). That the AI developers were under contract to take the data does not weaken the moral force of the act.

Can AI developers transcend this role-relative moral quandary? For the answer to this question, we turn to Alasdair MacIntyre (1999) who has formulated three criteria necessary for the exercise of professional moral agency. First, practitioners must understand themselves to be more than just the role they fill. He argues that “the powers of moral agency can only be exercised by those who understand themselves as moral agents” (p. 320). This means that one does not leave one's moral self at the office door or separate one's goodness as a human being from one's technical ability to fulfill a job function. More to the point, MacIntyre argues that people who see themselves as full moral agents will not agree to being a certain kind of person in one environment and another kind of person when the role calls for it.

Second, because practitioners will always be in potential or actual conflict with the requirements of their role, they must be able to justify their adherence to (or defiance of) those requirements on moral grounds. The confidence needed to make moral choices thus requires some experience with subjecting the reasons for one's choices to critical scrutiny. This means cultivating relationships of a particular kind; a community of “reliable others” to whom practitioners can “test [their] capacity for moral deliberation and judgment in this and that type of situation” (MacIntyre 1999, p. 316). These deliberations within a trusted and skilled community are what give practitioners rational confidence in their moral decisions. Without them, practitioners have “a seriously diminished type of agency, one unable to transcend the limitations imposed by [their] social and cultural order” (MacIntyre 1999, p. 317).

Finally, there must be an environment in which exercising moral agency is possible. Rational moral agents must be able

¹ At the time of this writing, at least two dozen lawsuits have been filed against technology companies for copyright infringement related to how they procured the data to train their generative models. We contend that the act of taking proprietary information without consent was immoral even if the law decides in favor of the technology companies. That something is deemed legal does not make it moral, nor are moral actions always legal ones.

to make decisions in an environment that does not mislead or deceive them. This requires an organizational culture that values transparency, trust, and open discussion about the work moral agents undertake. It is only in an environment where people can “understand themselves as accountable to others in respect of the human virtues and not just in respect to their role-performances” that they can act as full moral agents (MacIntyre 1999, p. 321).

Meeting all three of these requirements has been a notoriously insidious problem in AI development, and indeed across most of computer science, because of the problem of many hands. We turn now to an explanation of the problem of many hands, how it manifests in AI development, and the excuses it produces to justify leaving AI developers out of our moral deliberations.

3 Rethinking the problem of many hands

For the last four decades, scholars working in and around organizational and professional ethics have wrestled with “the problem of many hands” (Bovens 1998; Mastop 2010; Nissenbaum 1996; Thompson 1980, 2017; Van de Poel 2015). The term was first conceptualized by Thompson (1980) to describe the difficulty in assigning blame for detrimental public policies because an array of civil servants and bureaucrats was involved in their enactment. He observed that “many political outcomes are the product of the actions of many different people whose individual contributions may not be identifiable at all, and certainly cannot be distinguished significantly from other people’s contributions” (p. 907). Thompson was concerned about this diffusion of responsibility, and he was unimpressed with the excuses used to legitimize the collective shrug that would follow incidents of public harm. Thompson’s concern was later echoed by Bovens (1998), who observed that if no one could be held meaningfully accountable for an action after the fact, then “no one need feel responsible beforehand” (p. 49).

It was Nissenbaum (1996) who first assessed the problem of many hands in the context of computing and software engineering. She observed that computer systems are made by (sometimes large) groups of programmers, working in diverse settings, using stackable modules that have varying degrees of compatibility. As was the case then, and is now, developers rarely build systems from scratch, and their work is often re-packaged, marketed, and deployed in a multitude of scenarios. Nissenbaum observed that while software engineers can work independently, they are still part of a collective of other engineers, and a key feature of this collective is that no one agent is distinguishable from another, so none of their contributions can be singled out as the cause of harm.

Both Thompson (1980) and Nissenbaum (1996) identified several “excuses” that are used to legitimize not holding

anyone accountable for harms arising from a product or outcome produced by many hands. Thompson coined the excuses of *intentionality*, *ignorance*, and *compulsion*. Nissenbaum adds the excuse of *inevitability*.² These excuses are relevant to the arguments in this article because they are being used by AI developers. Evidence for this is presented in Table 1, which we gathered from a structured literature search of studies that engaged AI developers about their agency and perceptions of responsibility.

In what follows, we briefly describe each excuse and why it fails to free developers from their moral agency. It is important to note that in many cases, just as one excuse begins to fail, another is called upon. There can thus be some overlap between the excuses; they do not always flow iteratively and can be invoked in isolation or in combination. As we will show, whenever any of these excuses are invoked, the onus of moral responsibility for what the developer makes is delegated to anyone other than the developer.

We begin with the excuse from *intentionality*, in which practitioners claim they can only be held responsible for what they intended to build, not what anyone else did with the product afterwards (Bartsch et al. 2024; Griffin et al. 2023; Horváth and Vicsek 2023; Nichol et al. 2023). This excuse has a certain intuitiveness to it. After all, developers are hired to address a specific use case. They cannot control what other people do with a product after it is sold or incorporated into another system. In our review of the literature, we do find several examples of developers relying on their intentions as a means of excusing themselves from moral responsibility. Some of these excuses are more cynical than others. In a study by Horváth and Vicsek (2023), for example, one developer claimed that engineers are only interested in the creative challenge, stating, “Engineers who develop such facial recognition algorithms are not responsible for where they will be used and if they will discriminate. I think a lot of engineers see this [type of work] as a challenge—to see if they can build it, and if they can, [the job is] done” (p. 612). Most examples, however, were more functional in nature, with developers claiming that the technology they create is intended to be a blank slate, a morally neutral device that takes on moral valence only in the hands of an end user.

This rationale is a logical trick. Not only is the technology not neutral, but the developers relying on this excuse implicitly admit that harmful uses are possible and even foreseeable. The appeal to intentionality is, therefore, not

² Nissenbaum’s identified excuses are: bugs, the computer as scapegoat, and ownership without liability. We have combined Nissenbaum’s first two into the excuse from *inevitability*. The third, ownership without liability, does not directly relate to the role of computing professionals, but rather to the role of corporate managers and lawyers, and so we do not engage with it in this article.

Table 1 Examples of excuses arising from the “problem of many hands” in AI development milieus

Examples of excuses arising from the “problem of many hands” in AI development			
Categories derived from Thompson (1980)			Category derived from Nissenbaum (1996)
Intentionality	Ignorance	Compulsion	Inevitability
Developers believe that personal consequences will only occur if they have intentionally caused a mistake. Unintentional mistakes are not seen as problematic by developers because they are human (Bartsch et al. 2024, p. 5832)	For example, whenever [one participant] would discuss ethical issues, they would bring up the fact that they were engineers and that the engineering culture would usually ignore these issues and pretend that the problem was purely objective (Barbosa et al. 2023, p. 7)	[...] developers see themselves as exposed to the specifications given by others and, therefore, do not perceive algorithmic accountability (Bartsch et al. 2024, p. 5830)	[...] developers accept that someone needs to be algorithmically accountable, but they do not want to justify every single error or bug of their AI-based systems. Instead, they request that the public take a more nuanced view [...] because, by technical design, [the systems] are based on probabilities and can produce false outcomes (Bartsch et al. 2024, p. 5831)
“We don’t consider ourselves to be nefarious agents. It’s not that we have the pinnacle of high ethical standards, but we are also not evil geniuses sitting in our lair trying to figure out ways to hurt people” (Griffin et al. 2023, p. 4)	Human-out-of-the-loop systems contrast with human-in-the-loop systems, where developers locate the algorithmic accountability not with themselves but with the user, who is supposed to scrutinize AI-based system’s suggestions before using them (Bartsch et al. 2024, p. 5831)	[One practitioner argued there was a] “protective- defensive type of attitude” in which some developers believe “it is for the greater good that the model be out there [...] and anything that’s slowing that down is like, ‘Is it ethical to keep this internally much longer when it could be out in the world to save lives?’” (Griffin et al. 2024, p. 8)	[...] if unemployment was mentioned, it was seen mainly as a necessary inconvenience on the road to a better future for most members of society. Only in one interview was it stated that it would be the responsibility of the wealthy and better educated to help those who were losers of the changes (Horváth and Vicsek 2023, p. 611)
“Now, I’m sorry, the Swiss knife maker doesn’t care what his knife is used for, even whether it’s used for killing, to some degree. Engineers who develop such facial recognition algorithms are not responsible for where they will be used and if they will discriminate. I think a lot of engineers see this [type of work] as a challenge—to see if they can build it, and if they can, [the job is] done.” (Horváth and Vicsek 2023, p. 612)	Moreover, because many of the developers interviewed for this project perceive these technologies as neutral, or as “just tools,” the onus for ethics was situated outside the occupation. Ethics was instead located in the use case, and with the users (Griffin et al. 2023, p. 5)	“We don’t have the luxury of saying, ‘Oh, we are supporting this particular locale and this particular language in this particular circumstance.’ No, no, no, we’re doing it all! We’re doing it all at once, and we are being asked to ship more, faster. I mean, that is the pressure [...] and there will be tension for anything that slows that trajectory, because the gas pedal is to the metal” (Madaio et al. 2022, p. 11)	Some even expressed resignation or inevitability of conflicting interests leading to misuse. “[...] and these are not things that I advocate nor does the entire...our company advocate at all, but...at the end of the day a company’s gonna do what a company’s gonna do” (Nichol et al. 2023, p. 6)
Some of the harms of ML were recognized but were either believed to be justified by benefits, minimized by being characterized as being irrelevant to the interviewee’s work product, or by downplaying the role of ML in health care, usually by locating ultimate decision-making authority with a clinician (Nichol et al. 2023, p. 4)	Responsibility for the outcomes of AI systems [is] thus partially placed with the users who engage them. [...] “We were a technology provider, so we didn’t make those decisions...It is the same as someone who builds guns for a living. You provide the gun to the guy who shoots it and kills someone in the army, but you just did your job and you made the tool” (Orr and Davis 2020, pp. 729–730)	“Data scientists take on what they think is the ethical framework that they are supposed to be operating under, and so it is usually, how do we maximize profits? How do we maximize user engagement, user retention? How do we make this game the most addictive, how do we get more people to sign up?” (Orr and Davis 2020, p. 727)	Indeed, our data reveal a set of accountability norms which preclude simple calculus about blameworthiness should systems go awry—and given recent history with AI, ‘going awry’ is near inevitable (Orr and Davis 2020, p. 732)

The method for the literature review involved structured searches in Scopus, ACM Digital Library, Web of Science, and Google Scholar for abstracts with the following search criteria: (“artificial intelligence” OR “machine learning” OR “data science”) AND (ethic* OR moral*) AND (“developer” OR “programmer”) NEAR (interview*). Abstracts were then reviewed to include only studies in which AI developers spoke about their agency, decision-making processes, or perceptions of accountability. This resulted in 12 studies, which we closely reviewed for evidence that the developers interviewed were relying on any of the excuses from Thompson and Nissenbaum

Each entry in this table represents verbatim text from the cited source. It is important to note that the authors cited are not making these claims. They are, rather, reporting on claims they heard from their research participants. Quotation marks are used exclusively for statements made by study participants rather than the authors’ prose. Ellipses not contained in brackets are native to the cited source; content in brackets ([...]) indicates that text was added or removed by us to aid in brevity

a sufficient reason to free a practitioner from responsibility for taking precautions against such harmful uses. That “I did not intend for it to be used this way” may be true (and even forgivable). But it does not then follow that developers ought to just carry on limiting their moral responsibility to the immediate task. They are using their technical imagination to design, build, and deploy powerful pieces of technology. Competent practitioners can be expected to also use their moral imagination to, at a minimum, anticipate nefarious or reckless secondary uses and implement reasonable safety measures to prevent or limit harm.

Of course, it is difficult, if not impossible, to anticipate every possible use for an automated system. This is typically when the excuse from *ignorance* is called upon. Practitioners will claim they did not know, or could not be expected to know, all of the uses to which their creations could be put (Barbosa et al. 2023; Bartsch et al. 2024; Griffin et al. 2023; Kasun et al. 2024; Orr and Davis 2020). On the basis of the excuse from ignorance, practitioners involved in designing, developing, or deploying algorithmic systems feel (and are permitted to feel) a diminished sense of responsibility. They may claim that the “black box” nature of some systems leaves them just as in the dark as everyone else; or that the systems are “just tools” that have myriad uses, good and bad, that are unknowable to them (Griffin et al. 2023). In this way, they again shift the responsibility for the harmful effects of their own software code to other agents, typically the end user (Bartsch et al. 2024; Griffin et al. 2023; Orr and Davis 2020).

This excuse also encourages practitioners to become, in Thompson’s (1980) words, “instruments of their own ignorance” (p. 912). They can willfully avoid learning anything about the uses of the systems they build or ask to be kept in the dark. This is illustrated by one participant interviewed by Barbosa et al. (2023), who noted that “engineering culture would usually ignore [ethical] issues and pretend that the problem was purely objective” (p. 612). But ignorance can only be a legitimate justification when obtaining necessary information for ethical decision-making would be an unreasonable task. One cannot expect a nurse who is tasked with injecting a particular medication to first find out everything about how that medication was manufactured to make sure the factory did not err. But AI developers are exactly the experts who are at this manufacturing level. They have access to the relevant data, select what data to use, create the algorithm, and verify its outputs.

Third is the excuse from *compulsion*, wherein developers claim they are simply doing their job and had no other choice (Bartsch et al. 2024; Griffin et al. 2023; Madaio et al. 2022; Orr and Davis 2020; Widder et al. 2023). Here, developers bypass the question of whether they could have made a difference in the outcome at all. The excuse applies in situations where the developer’s superiors demand that they move

faster than the pace of ethical reflection would allow or not provide enough support to make ethics evaluations part of the workflow.

But this escape from responsibility ignores the fact that developers have a great deal of discretionary authority, both delegated and assumed. Take, for instance, the following verbatim from a data scientist interviewed by Nichol et al. (2023, p. 5):

“Now, a data scientist has tremendous powers here because your stakeholders don’t really understand what precision recall is and where that threshold should be, so it’s up to you to use your own judgment and say, you know what, actually I think I would rather that people have their claims paid than denied, so I will just tune it for true-positives”—Participant 24.

This statement from Participant 24 is remarkable not only for the developer’s awareness of their own technical skill, but also the fact that their level of knowledge empowers them to make ethical decisions without anyone knowing about them. This illustrates that developers do recognize their authority to act and will admit to wielding that authority when they believe the situation calls for it. Several developers interviewed by Widder et al. (2023), for example, revealed that they would purposefully do a poor job when a superior asked them to produce a product the developer thought was unethical. One went so far as to lament their decision to not take a job on ethical grounds, reflecting that they could have done a bad job of it, but instead freed the company to find someone with a weaker moral compass (Widder et al. 2023).

What Nissenbaum (1996) adds to this discourse is what we call the excuse from *inevitability*. When leaning on this justification, developers will call upon three pretexts: bugs, the computer as scapegoat, or the inescapability of progress (Bartsch et al. 2024; Griffin et al. 2023; Horváth and Vicsek 2023; Madaio et al. 2022; Nichol et al. 2023; Orr and Davis 2020). As we saw in the excuses from intentionality, ignorance, and compulsion, responsibility for the inevitable is offloaded onto others. Developers interviewed by Bartsch et al. (2024), for example, accept “that someone needs to be algorithmically accountable...” But in the same sentence, they also insist that it would be unfair to expect developers to take responsibility for “every single error or bug of their AI-based systems” (p. 5831). Instead, they ask the public to take a more nuanced view because the probabilistic systems are guaranteed to produce false outputs (i.e., the computer did it). Similarly, some developers rationalize the consequences of automation, like unemployment, as a “necessary inconvenience on the road to a better future” and place the responsibility for mitigating that harm on wealthy or more educated people (Horváth and Vicsek 2023, p. 611).

The excuse from inevitability fails for the same reason as the excuse from intentionality fails: The fact that developers

know these issues are certainties makes them more, not less responsible because they have the proximity and the skill to resolve or at least mitigate them. And if mitigation is not within their sphere of influence, then they still can bring the problems they foresee to the attention of those who can mitigate them. Rather than invoking “many hands” as an excuse to free oneself from moral culpability, many hands should instead be relied upon to solve the problem.

What we detailed in this section is that even if practitioners did not intend harm, were ignorant of adversarial secondary use, were under compulsory pressure to finish the task without resistance, and/or felt justified in deferring responsibility to the inevitabilities of bugs or progress, they are still morally entangled in the outcomes of the systems they build. They are moral agents and are thus subject to moral standards, just as much as any others who had a hand in the creation or deployment of the end product. Or, as Applbaum (1999) succinctly puts it, “There is no law of conservation of moral agency, such that if an act is attributable to you, it cannot be attributable to me as well” (p. 248).

4 How they understand themselves

We have so far established that, from a theoretical point of view, AI developers are moral agents. We have also dispensed with the most common excuses for not taking that moral agency seriously. Our attention now turns to whether and how AI developers can contend with this moral agency. Returning to MacIntyre’s moral agency framework, we now ask: (1) do AI developers understand themselves as having a moral identity within their work role that is compatible with their moral identity as persons outside of their work role; (2) do they understand themselves as rational agents working in community with others to whom they can submit their moral choices for critical scrutiny; and (3) are they working in environments that are conducive to their moral actualization?

To answer these questions, we turn again to the available literature we used in the previous section. A summary of their reflections is presented in Table 2. We caution that the authors of these studies did not use MacIntyre’s framework in conducting their analyses. Ours thus presents a “best fit” along each of these three axes. Our review reveals that many (but not all) developers do think of their moral agency as something that transcends their work role. However, their community of “reliable others” is inconsistent and bounded by time and resource constraints. Likewise, their ability to appropriately exercise this agency, both individually and collectively, is often a matter of their environment, which is limited by resource constraints, company culture, and transparency.

4.1 Perceived moral agency

Starting with MacIntyre’s first criterion, the evidence from the studies we analyzed reveals that many developers do think of themselves as having a moral identity beyond the role they are paid to fill. We categorized the responses along this axis as *strong*, *constrained*, or *absent* based on the level of assurance the respondents seemed to have.

When developers expressed a *strong* sense of moral agency, it was when what they considered to be right and wrong were clear to them. We have already discussed one example of this from Participant 24 in the Nichol et al. (2023) study, who tuned a health insurance algorithm for true positives because the developer “would rather people have their claims paid than denied.” Other developers have also stated that they will purposefully do a poor job or will explicitly refuse to comply if they are asked to do unethical work (Widder et al. 2023). Some developers thus exercise their moral agency without hesitation and even without permission when they think the situation calls for it. Importantly, developers in one study also expressed a strong sense of responsibility for the technomoral aspects of the work, such as meeting robustness standards, obtaining end-user consent, and even preparing for the inevitable misuse of a product (Howes et al. 2024, p. 12).

The evidence that at least some developers understand themselves as moral agents also shows up when the stakes are more nuanced. Responses were categorized as *constrained* when the developer reflected that their capacity to mitigate a known harm was not as straightforward, although they did recognize their agency. In the Nichol et al. (2023) study, for example, one practitioner noted, “Say the treatment for a certain indication puts somebody under a lot of duress, and if we falsely flag somebody as having that indication then the culpability of that duress, you know, at least partially does lay on our shoulders” (p. 5). Another developer interviewed for the Widder et al. (2023) study experienced severe moral dissonance with the demands of the role saying, “I was so distraught over what I was being asked to do, I threw up in the parking lot before going to work” (p. 472). This dynamic tension between the developers’ expertise, moral sensitivity, and their obligations to a client or superior led Orr and Davis (2020) to observe that the developers in their study “cast themselves as constrained autonomous agents, encoding others’ agendas while owning the ethical responsibility that accompanies technical processes of making” (p. 729).

Notwithstanding these examples of affirmative identification with moral agency, in some contexts, developers are still quite apathetic, and here we coded their responses about moral agency as *absent*. Developers interviewed by Horváth and Vicsek (2023), for example, “had not really thought before about the effect of AI on society” and almost

Table 2 Examples of developer perceptions of moral agency, moral community, and moral environment

Examples of developer perceptions of moral agency, community, and environment
Categories derived from Macintyre (1999)

Moral agency	Moral community	Moral environment
Strong One practitioner who was asked to build a system to bypass spam filters wrote: “I purposefully created a poor implementation and did not dedicate very much energy to make a working solution.” Another respondent wrote that they “pretended to complete the task but didn’t” (Widder et al. 2023, p. 471)	Conditional [...] developers tend not to have quality management performed by external or internal departments but to integrate it into their team of developers. In this manner, developers ensure that sufficient technical knowledge about AI-based systems is available and that errors and bugs within AI-based systems are more likely to be discovered and corrected (Bartsch et al. 2024, p. 5830)	Conducive [...] developers who work in established organizations are more likely to perceive algorithmic accountability because they receive more organizational support (Bartsch et al. 2024, p. 5831)
Strong “I looked that manager in the eye and I said: you are going to have to write me up or fire me, but I’m not doing it. I’m not going to put patients’ lives at risk [just] because you’ve got a pile of money sitting on the table.” This confrontation resulted in management stepping back and reassessing the necessity of the update (Widder et al. 2023, p. 471)	Conditional We found that [the conversational approach between developers] made them recognize their own responsibility for the artifact and its eventual consequences, deviating from standard practice in computing documentation where these sorts of statements are depersonalized and presented as more objective statements (Barbosa et al. 2023, p. 7)	Conducive “We have policy people [...] [who work with] AI researchers from philosophy, ethics, and social science backgrounds [...] [Their] guidance documents are what we follow internally. Now, we can challenge this. There is a healthy debate internally, but if you are rushed, you will just follow whatever the policy guidance says. So, we basically rely on those teams being good” (Griffin et al. 2024, p. 7)
Constrained [Developers’] expert status makes them indispensable and at the same time makes a degree of personal control and autonomy unavoidable. For these reasons, participants cast themselves as constrained autonomous agents, encoding others’ agendas while owning the ethical responsibility that accompanies technical processes of making (Orr and Davis 2020, p. 729)	Conditional [When there is transparency about a project] developers more often ask team members for a second opinion as reassurance and integrate it into their development. Simultaneously, developers communicate more openly about their problems and challenges [...] (Bartsch et al. 2024, p. 5833)	Restricted “You have to take everything in its context and have to be very careful to try to understand that the data generation process [...] the doctor is going to do whatever they believe is best for the patient, and they’re going to play the game to make that happen in a lot of cases [...] We then play with that data and pretend that it’s really clean and useful for building [a model]. We need to be very, very careful about that” (Kasun et al. 2024, p. 568)
Constrained “Say the treatment for a certain indication puts somebody under a lot of duress, and if we falsely flag somebody as having that indication then the culpability of that duress, you know, at least partially does lay on our shoulders” (Nichol et al. 2023, p. 5)	Limited The likelihood of ethics negotiations succeeding are, as one practitioner put it, “entirely [dependent] on the organization and your ability to talk to people and [...] capture hearts and minds” (Widder et al. 2023, p. 471)	Restricted [...] decisions about performance metrics are shaped by business imperatives that prioritize some stakeholders (e.g., customers) over others (e.g., marginalized groups). Participants described how tensions that arose during discussions about choosing performance metrics [...] were often indicative of deeper disagreements about the goals of their AI systems. These tensions were exacerbated by a lack of engagement with direct stakeholders or domain experts (Madaio et al. 2022, p. 529)

Table 2 (continued)

Examples of developer perceptions of moral agency, community, and environment
Categories derived from Macintyre (1999)

Moral agency	Moral community	Moral environment
Absent “I don’t think there’s a general ethical question even. We don’t think of ourselves as unethical, but only because we’re not wondering if what we’re doing is ethical or not”—7015_ArtificialIntelligence (Griffin et al. 2023, p. 4)	Limited Another survey respondent suggested that raising concerns could lead to hostility: “I do not want to judge, or be judged, by colleagues for my views. Without care, such discussions can lead to a hostile work environment” (Widder et al. 2023, p. 472)	Non-conductive “There are few places you can go for help. Some things you have to escalate to the top level [to] force people on the product side to at least give you access so you can show there’s a problem. It’s very difficult to raise an ethics question without having any kind of proof; it’s very difficult to have proof without access, and all access is regulated” (Griffin et al. 2024, p. 7)
Absent There was almost unanimous agreement about the idea of the developer as a tool without any real capacity to shape the direction of innovation or to influence the misuse of technology (Horváth and Vicsek 2023, pp. 611–612)	Limited One data science practitioner who was volunteering time with the responsible AI team shared, “More senior people are making the decisions. I saw ethical concerns but there was difficulty in communicating between my managers and the [responsible AI] team. People weren’t open for scrutinization” (Rakova et al. 2021, p. 14)	Non-conductive For example, [two] interviewees [...] came up with technical solutions that would have resolved their concerns, [but] which were dismissed by management (Widder et al. 2023, p. 470)

The studies cited in this table are the same studies we used to populate Table 1. The method for the literature review involved structured searches in Scopus, ACM Digital Library, Web of Science, and Google Scholar for abstracts with the following search criteria: (“artificial intelligence” OR “machine learning” OR “data science”) AND (ethic* OR moral*) AND (“developer” OR “programmer”) NEAR (interview*). Abstracts were then reviewed to include only studies in which AI developers spoke about their agency, decision-making processes, or perceptions of accountability

Each entry in this table represents verbatim text from the cited source. It is important to note that the authors cited are not making these claims. They are, rather, reporting on claims they heard from their research participants. Quotation marks are used exclusively for statements made by study participants rather than the authors’ prose. Ellipses not contained in brackets are native to the cited source; content in brackets ([...]) indicate that text was added or removed by us to aid in brevity

unanimously thought of themselves as “a tool without any real capacity to shape the direction of innovation or to influence the misuse of technology” (pp. 611–612). Additionally, two participants in the Griffin et al. (2023) study said they actively avoided thinking about their ethical responsibilities, with one noting that they preferred “learning and trying to implement AI, not thinking about ethics” (p. 4).

4.2 Perceived moral community

Turning now to MacIntyre’s second criterion, we investigate whether developers understand themselves as members of a community of rational practitioners to whom they can safely submit their choices for ethical scrutiny. Developers clearly desire to be part of a strong moral community, as expressed by developers across multiple studies (Bartsch et al. 2024; Griffin et al. 2024; Horváth and Vicsek 2023). In fact, the same practitioners who thought of themselves only as tools also spoke of “the need to bring a more philosophical way of thinking to their workplaces and expressed a longing for their university years when they had the opportunity to talk about such issues” (Horváth and Vicsek 2023, p. 612). But that does not mean they also have access to such a community. Indeed, in reviewing the literature, it was not apparent that *any* developers had reliable access to a robust moral community. Our categorization of responses hence fell into only two categories: *conditional* and *limited*.

Where we noted that developers’ access to a community of reliable others was *conditional*, it was often based on their proximity to decision-making authority. The Bartsch et al. (2024) study, for example, found that practitioners were more likely to accept accountability if quality management were housed within the team, while Griffin et al. (2024) found a strong preference for ethical wisdom to be strengthened via internal team structures rather than from outside trainings. Likewise, Barbosa et al. (2023) found that when developers spent time together talking through the stages of development, they were more likely to recognize their own responsibility for the technology and its eventual consequences. This stands in contrast to the standard practice of documenting design and deployment choices in writing, which often prioritizes depersonalized and objective statements, rather than reflective and personal ones.

In addition to the conditional nature of developers’ moral communities, several also mentioned that their ability to access the moral wisdom of others was *limited* by their own sense of personal and professional safety. Some noted that they would “more often ask team members for a second opinion as reassurance and integrate it into their development” if there was transparency about a project from the start (Bartsch et al. 2024). Still others were acutely aware that the likelihood of ethics discussions succeeding was “entirely [dependent] on the organization and your ability

to talk to people and [...] capture hearts and minds” (Widder et al. 2023). Another respondent in this same study worried that raising concerns could lead to hostility: “I do not want to judge, or be judged, by colleagues for my views. Without care, such discussions can lead to a hostile work environment” (Widder et al. 2023).

4.3 Perceived moral environment

MacIntyre’s third criterion requires that the practitioners’ environment is conducive to their moral actualization and does not deceive them. In reviewing the literature, we do find evidence that developers recognize the conditions that are *conducive*, those that are *restricted* by other factors, and those that are all together *non-conducive* to their moral decision-making confidence.

Developers understood that for an environment to be *conducive* to their moral functioning, they need organizational support and time for reflection, which they recognize is available in more mature organizations, but not at startups (Bartsch et al. 2024). Developers also recognized that having a common goal and open-door policies with senior leaders in the organization makes it easier for them to exercise their moral agency (Widder et al. 2023).

Nevertheless, many studies we reviewed revealed that practitioners’ ability to act as moral agents is often *restricted* by business incentives. These incentives prioritize performance metrics over ethical reflection or considering the downstream impacts of an AI deployment (Griffin et al. 2024; Madaio et al. 2022; Nichol et al. 2023; Orr and Davis 2020; Widder et al. 2023). In the Madaio et al. (2022) study, for example, participants described how tensions arose when business interests required prioritizing customers over marginalized groups. These tensions were “exacerbated by a lack of engagement with direct stakeholders or domain experts that might be harmed by the system” (p. 529). In this way, the commercial ground upon which AI developers build their systems often encourages them to restrict their practical reasoning in order to discharge their responsibilities to the business.

Importantly, the developers’ ability to exercise their moral agency is not only dependent on workplace culture but is also heavily reliant on the community of data generators. Consider, for example, the developer working on healthcare applications in contexts where physicians are inputting data that may not be accurate:

“You have to take everything in its context and have to be very careful to try to understand that the data generation process [...] the doctor is going to do whatever they believe is best for the patient, and they’re going to play the game to make that happen in a lot of cases [...] We then play with that data and pretend that it’s

really clean and useful for building [a model]. We need to be very, very careful about that” (Kasun et al. 2024).

Developers are thus required to maintain a constant awareness of how the data generator could be deceiving them and under what conditions, knowing that everything they build will thus be perpetuating that first deception.

The environments that are all together *non-conducive* to moral reasoning and action were those in which the developers’ concerns are simply not taken seriously. Two participants in the Widder et al. (2023) study, for example, said they had designed technical solutions that would have addressed their ethical concerns, but those solutions were dismissed by management. A participant in the Griffin et al. (2024) study similarly expressed distress over knowing what to do but being prevented from doing so, noting, “It’s very difficult to raise an ethics question without having any kind of proof, it’s very difficult to have proof without access, and all access is regulated” (p. 7).

5 Conclusion

We return now to where we began. AI developers are engaged in a practice of remaking, refining, and optimizing practices that are core to our societal functioning. The very nature of their work invests them with awesome powers over an increasingly dependent public. This fact alone requires that these practitioners take their own moral agency seriously. In making our argument, we placed Arthur Isak Applbaum, Alasdair MacIntyre, Dennis Thompson, and Helen Nissenbaum in conversation with AI developers about the rules of the AI developer role. We evaluated how and why excuses from “the problem of many hands” are invoked and showed that they are an insufficient justification for hiding from view the moral agency of **arguably the most powerful agents in the industry**. We then showed that AI developers do understand themselves as being responsible for more than just the role they fill yet are limited by the lack of a robust community to whom they can submit their choices for ethical scrutiny and **a work environment that is often non-conducive to their moral actualization**.

At a minimum, we argue that the AI industry and society at large must acknowledge that AI developers do not lose their freedom and moral sensitivity just because they took on an occupational role. They should be empowered and expected to engage as moral agents when they are at work. Our analysis shows that at least some of the developers in the field agree, with one admitting that they have a responsibility to “be sticks in the mud about ethics” because “this work doesn’t get done without us” (Widder et al. 2023, p. 472).

This admirable commitment notwithstanding it is also possible (and arguably probable) that many developers

see themselves as lacking moral agency or do not want to accept it at all. It may be easier to remain hidden behind the technical aspects of the work and leave it to others to clean up the messes that occur. But if the goal of ethics in AI is to prevent harm and heal relationships broken by systemic injustices, then we must be able to call upon the actors involved to take appropriate action. How to best organize developers to accomplish this, including what protections and restrictions should be in place via professionalization, regulation, or otherwise, is the subject of a different article. Our work here is only meant to establish that the status quo is morally unjustified. We do, however, see three important implications from the available research so far for professional ethics in the AI field.

First, **universities** and training institutions must **instill in developers a stronger sense of responsibility for the outcomes of their work**. Developers need to come out of these programs with a deeper understanding of themselves as moral agents who can identify the broader, morally relevant horizons of their work. Second, professional associations should require ethical reflection as a condition of publication or presentations within the membership. Conferences and papers dedicated solely to ethics are good and appropriate, but they should not be treated as secondary to the science of AI instead of integral to it. If the discipline is to mature as a moral one, its practitioners need to insist on submitting their work for both technical and ethical scrutiny as core to the tradition. Third, organizations employing AI developers, machine learning engineers, and data scientists should cultivate environments that do not deceive or manipulate developers’ ability to exercise moral agency. Given what is at stake in terms of societal functioning, these practitioners should be fully briefed on the use case and given access to relevant specialists. They should also be provided with other supports that aid in continuous moral decision-making, from the initial testing to refining, and especially the maintenance of automated systems (see, for example, Drage et al. 2024).

Ultimately, AI developers ought to take to heart what Applbaum (1999) bluntly concluded about people occupying roles that invite adversarial relationships with the public. Namely, that they “*are* responsible for the vice and stupidity of their trade and *should* refuse to practice [it] in vicious and stupid ways” (p. 260, emphasis in the original).

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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